San Francisco 49ers Offense: Scouting Report

Public data. Built from nflverse (nflfastR play-by-play, NGS participation, FTN charting), mapped onto the PFF GamePBP layout. Run concepts, PFF success codes, yards after contact and receiver alignment are not in this source; see *What nflverse cannot cover* at the end.

Sample: 19 games (Sep 07, 2025 to Jan 17, 2026) · 1158 offensive plays · Opponents: ARZ, ATL, CAR, CHI, CLV, HST, IND, JAX, LA, NO, NYG, PHI, SEA, TB, TEN
Source: SF_2025_nflverse_GamePBP.csv · Generated Nov 05, 2026 16:45 · Success = standard yardage definition · EPA = nflfastR model

59%	52%	+0.07	53%	50%
PASS RATE	NEUTRAL EARLY-DOWN PASS RATE	EPA / PLAY	SUCCESS RATE	3RD-DOWN CONVERSION

4%	8%
SACK RATE	EXPLOSIVE PLAY RATE

Key findings

TENDENCIES

- Passes on 59% of all plays and 52% of neutral early-down plays (1st/2nd down, outside 2-minute drills).
- QB alignment is a strong run/pass key: under center they run 66% of the time (n=522); in shotgun they pass 83% (n=610).
- Pistol is a real part of the menu: 26 plays, 35% pass.
- Base personnel is 11 (44% of snaps, 74% pass). 4 groupings reach 10+ plays.
- Tell: 3rd Down + Long (7-10) → pass 95% of the time (n=56, 95% lower bound 85%).
- Tell: Shotgun + 3rd Down → pass 89% of the time (n=190, 95% lower bound 84%).
- Tell: 3rd Down + Medium (4-6) → pass 92% of the time (n=63, 95% lower bound 83%).
- Tell: 11 + Shotgun → pass 86% of the time (n=390, 95% lower bound 82%).
- Run game is built on Middle and Left Guard: 41% of carries.
- Predictability: knowing personnel, alignment, down and distance lifts a defense's run/pass guess from 59% to 74% (leave-one-out).

STRENGTHS

- Most efficient run gap: Middle, +0.12 EPA/carry, 4.0 YPC, 61% success (n=125).
- Primary target: C. McCaffrey (#23) (HB), 24% target share, 113 catches for 1009 yards, +0.33 EPA/target, 55 first downs.
- Most efficient receiver: G. Kittle (#85) (TE), +0.65 EPA/target on 71 targets, aDOT 6.7, 4.4 YAC/rec.
- Lead back C. McCaffrey (#23): 72% of carries, 3.8 YPC, 49% success. Favorite gap: Middle.
- Most total value: G. Kittle (#85), 44.9 total EPA on 72 touches + targets (631 scrimmage yards).
- Deep shots (20+ air yards): 10% of targets, 44% complete, +0.35 EPA/target (n=59).

VULNERABILITIES

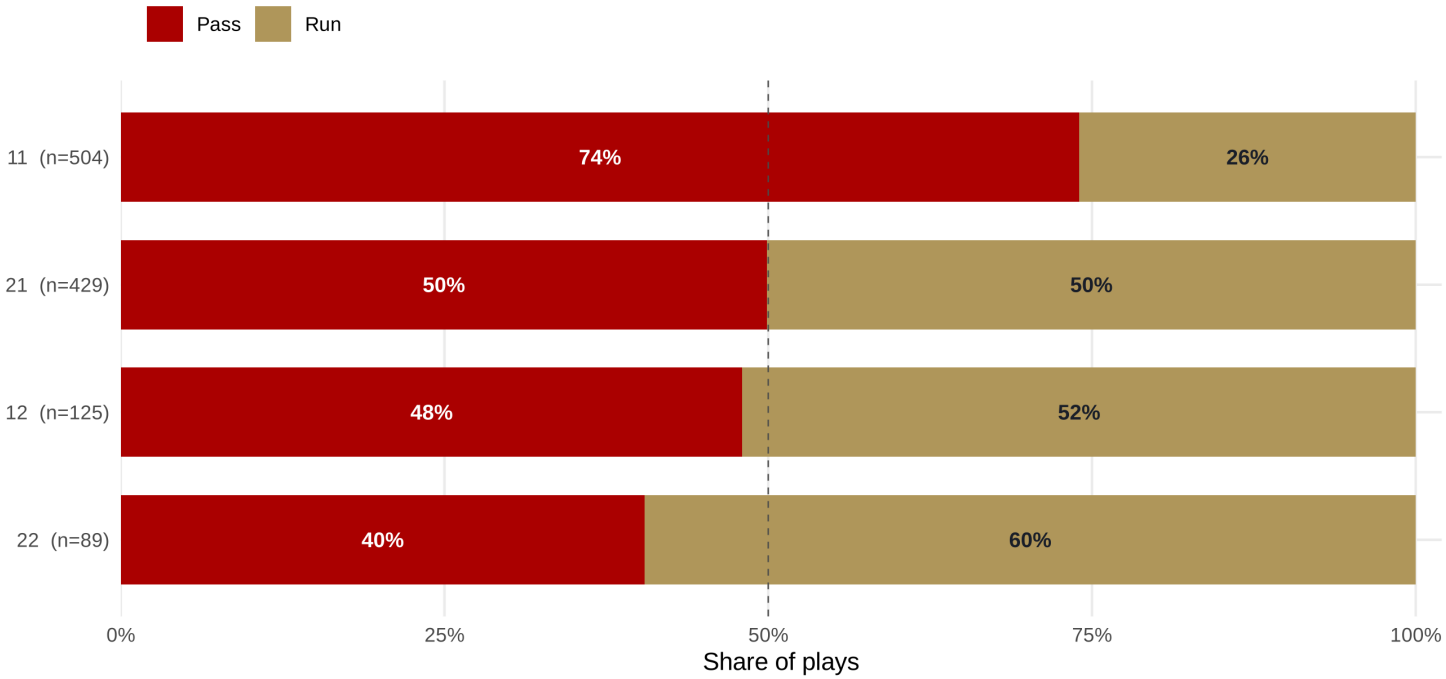
- Play action: 22% of dropbacks, +0.09 EPA/play vs +0.16 without (n=153 vs 536).
- Pressure breaks the passing game: -0.48 EPA/dropback when pressured vs +0.35 clean. Pressured on 25% of dropbacks; sack rate 4%.
- Least efficient run gap: Right End, -0.19 EPA/carry, stuffed (0 or fewer yards) on 25% (n=55).
- On 3rd and Extra Long (11+) they convert only 30% (n=33); they pass 91% of the time there. Overall 3rd-down conversion: 50%.
- Least productive personnel: 12, -0.03 EPA/play (n=125, 48% pass).
- High-volume, low-return target: J. Jennings (#15) (WR), 16% of targets but +0.00 EPA/target and 60% catch rate.

n = the number of plays (or carries / targets) behind a number, i.e. the sample size. Findings are generated by fixed rules (see Methodology) and include sample sizes. Treat any rate built on fewer than 10 plays as anecdotal.

1. Personnel run/pass tendencies

Run/pass split by personnel

Groupings with at least 10 plays. n = number of plays in that grouping



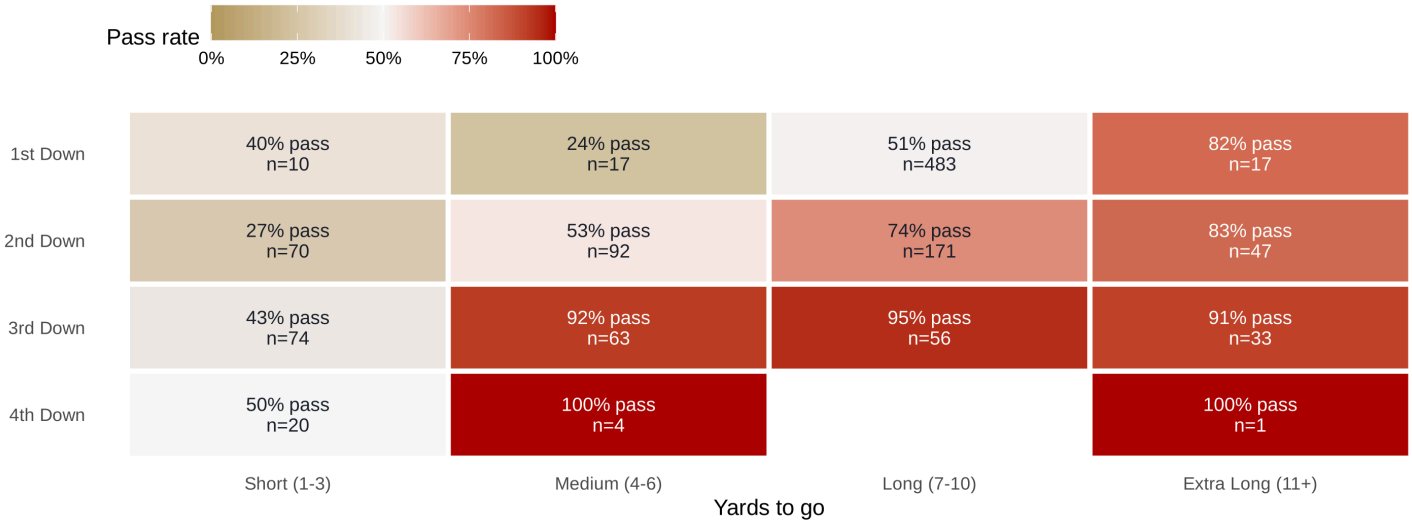
Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Personnel	Plays	Snap %	Pass %	Pass % 95% CI	Run EPA	Pass EPA	Success
11	504	44%	74%	70%-78%	-0.06	+0.14	48%
21	429	37%	50%	45%-55%	-0.04	+0.21	57%
12	125	11%	48%	39%-57%	-0.06	+0.01	53%
22	89	8%	40%	31%-51%	-0.00	-0.03	64%
23	9	1%	44%	19%-73%	-0.08	+0.40	56%
13	2	0%	100%	34%-100%	n/a	-0.25	50%

2. Down-and-distance tendencies

Pass rate by down and distance

Each box: share of plays that were passes, and n = number of plays in that down-and-distance situation



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Down	To go	Plays	Pass %	Run EPA	Pass EPA	Success
1st Down	Short (1-3)	10	40%	+0.05	+0.21	60%
1st Down	Medium (4-6)	17	24%	+0.18	+1.03	65%
1st Down	Long (7-10)	483	51%	-0.11	+0.04	55%
1st Down	Extra Long (11+)	17	82%	+0.09	+0.30	53%
2nd Down	Short (1-3)	70	27%	+0.01	+0.18	69%
2nd Down	Medium (4-6)	92	53%	+0.04	+0.16	60%
2nd Down	Long (7-10)	171	74%	-0.25	+0.11	43%
2nd Down	Extra Long (11+)	47	83%	-0.35	+0.33	40%
3rd Down	Short (1-3)	74	43%	+0.07	+0.26	64%
3rd Down	Medium (4-6)	63	92%	+0.22	+0.16	51%
3rd Down	Long (7-10)	56	95%	+0.57	+0.38	45%
3rd Down	Extra Long (11+)	33	91%	-0.05	+0.51	30%
4th Down	Short (1-3)	20	50%	+0.92	-0.18	65%
4th Down	Medium (4-6)	4	100%	n/a	-2.35	0%
4th Down	Extra Long (11+)	1	100%	n/a	-0.15	0%

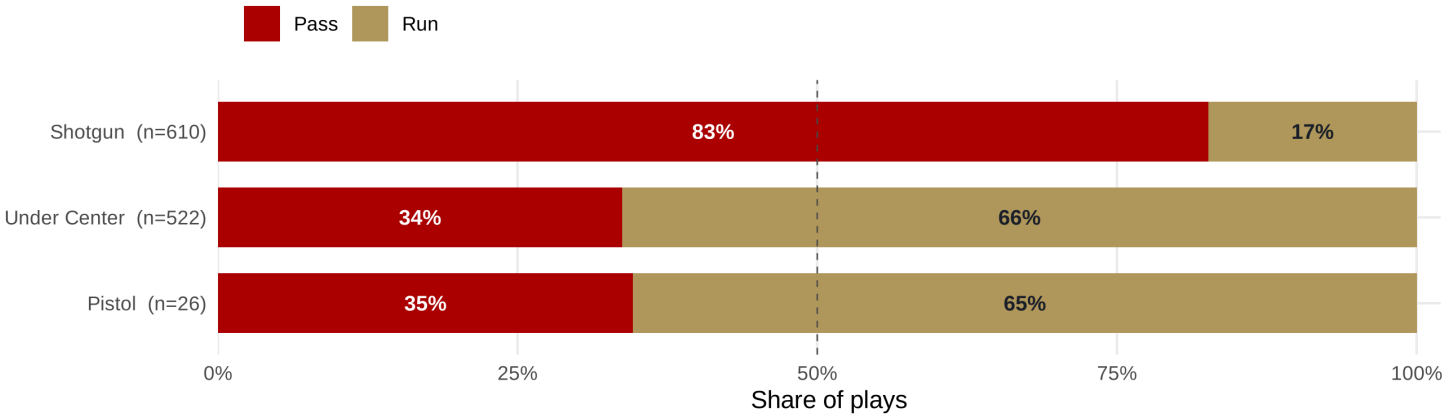
By field position (yards from the opponent's end zone):

Zone	Plays	Pass %	EPA/play	Success
Goal to Go (1-10)	96	53%	+0.31	56%
Red Zone (11-20)	110	51%	+0.10	46%
Plus Territory (21-50)	427	53%	+0.02	53%
Own Territory (51-80)	459	68%	+0.04	53%
Backed Up (81-99)	66	64%	+0.14	55%

3. QB alignment tendencies

Run/pass split by QB alignment

n = number of plays from that alignment; read small groups with caution



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Alignment	Plays	Snap %	Pass %	Run EPA	Pass EPA	Run YPC	Pass Y/P
Under Center	522	45%	34%	-0.03	+0.12	3.7	7.3
Shotgun	610	53%	83%	-0.10	+0.15	3.7	6.8
Pistol	26	2%	35%	-0.01	+0.37	3.9	9.7

Alignment by personnel (groupings with enough plays):

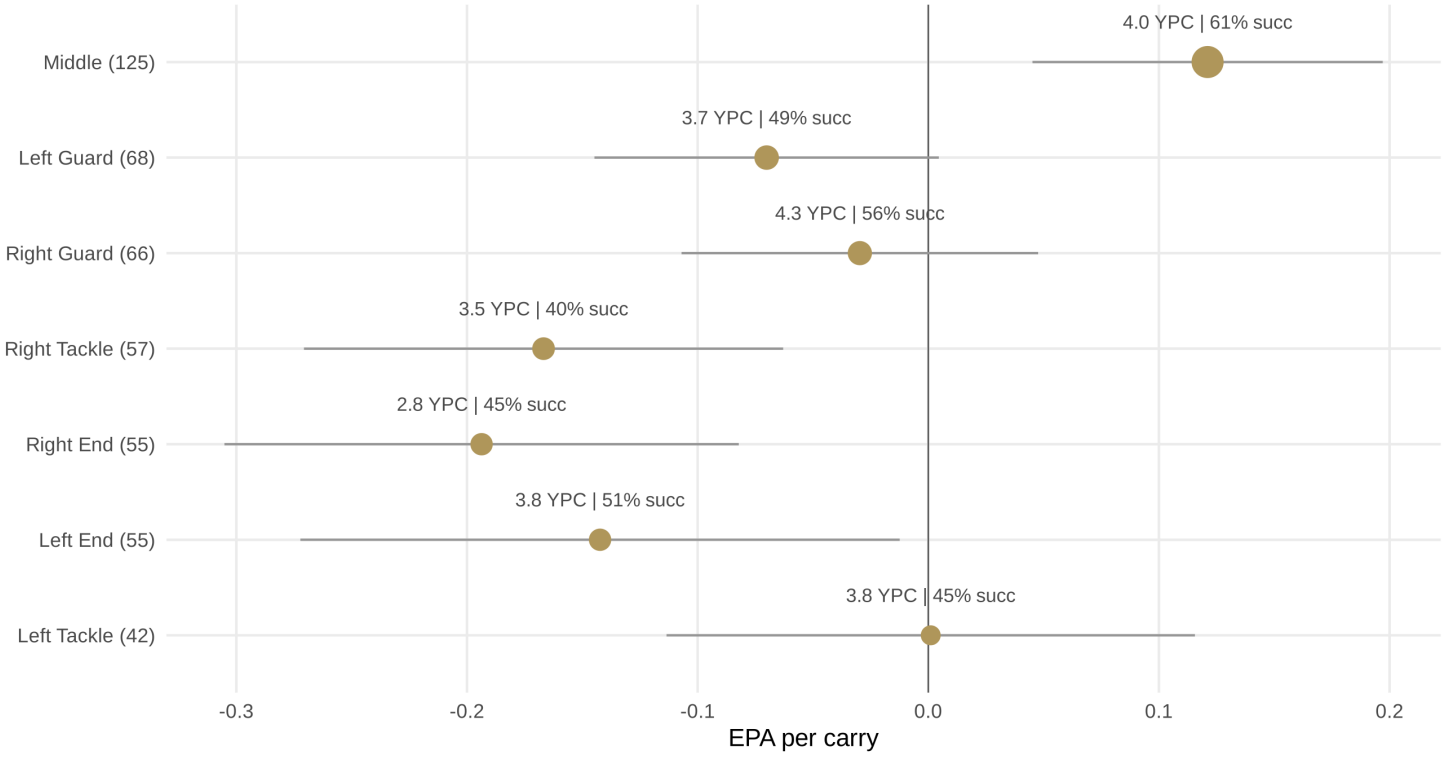
Personnel	Alignment	Plays	Pass %	EPA/play
11	Under Center	107	31%	-0.01
11	Shotgun	390	86%	+0.12
11	Pistol	7	71%	-0.24
12	Under Center	87	40%	-0.05
12	Shotgun	35	71%	-0.00
21	Under Center	252	34%	+0.10
21	Shotgun	163	77%	+0.05
21	Pistol	14	21%	+0.32
22	Under Center	67	27%	-0.13
22	Shotgun	20	85%	+0.38
23	Under Center	8	38%	+0.05

4. Run gap and direction: frequency and effectiveness

nflverse does not chart blocking schemes, so runs are grouped by the gap and side the play-by-play records (end / tackle / guard / middle). This answers *where* they run, not *which concept*.

Run gap and direction: volume and efficiency

Ordered by carries (in brackets). Point = EPA per carry, bar = +/- 1 standard error



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

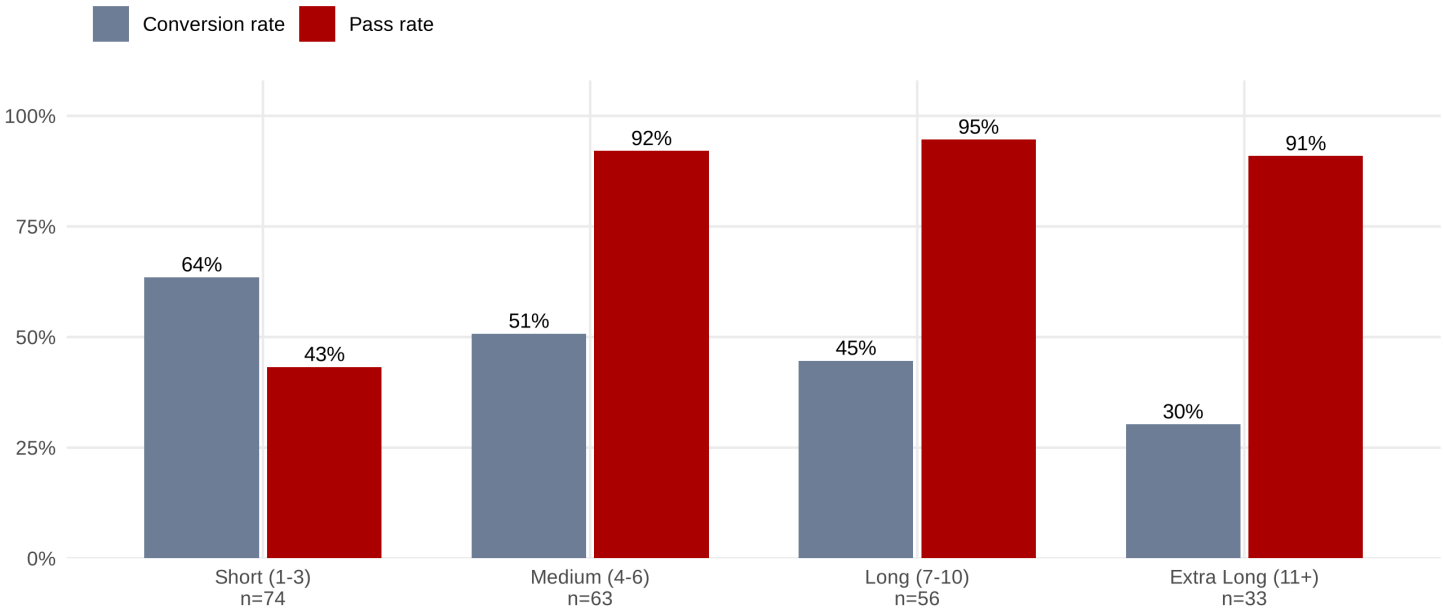
Concept	Carries	Share	YPC	Median	Success	EPA/carry	Stuff %	Explosive %	Under center %	Top personnel
Middle	125	27%	4.0	3.0	61%	+0.12	14%	10%	65%	21
Left Guard	68	14%	3.7	3.0	49%	-0.07	13%	4%	69%	21
Right Guard	66	14%	4.3	3.5	56%	-0.03	11%	8%	70%	21
Right Tackle	57	12%	3.5	2.0	40%	-0.17	26%	9%	86%	21
Left End	55	12%	3.8	3.0	51%	-0.14	24%	13%	78%	21
Right End	55	12%	2.8	2.0	45%	-0.19	25%	7%	78%	21
Left Tackle	42	9%	3.8	2.5	45%	+0.00	21%	7%	88%	21

1 carry with no charted run gap is left out of this table.

5. Third-down behavior and key targets

Third down: how often they throw and how often they convert

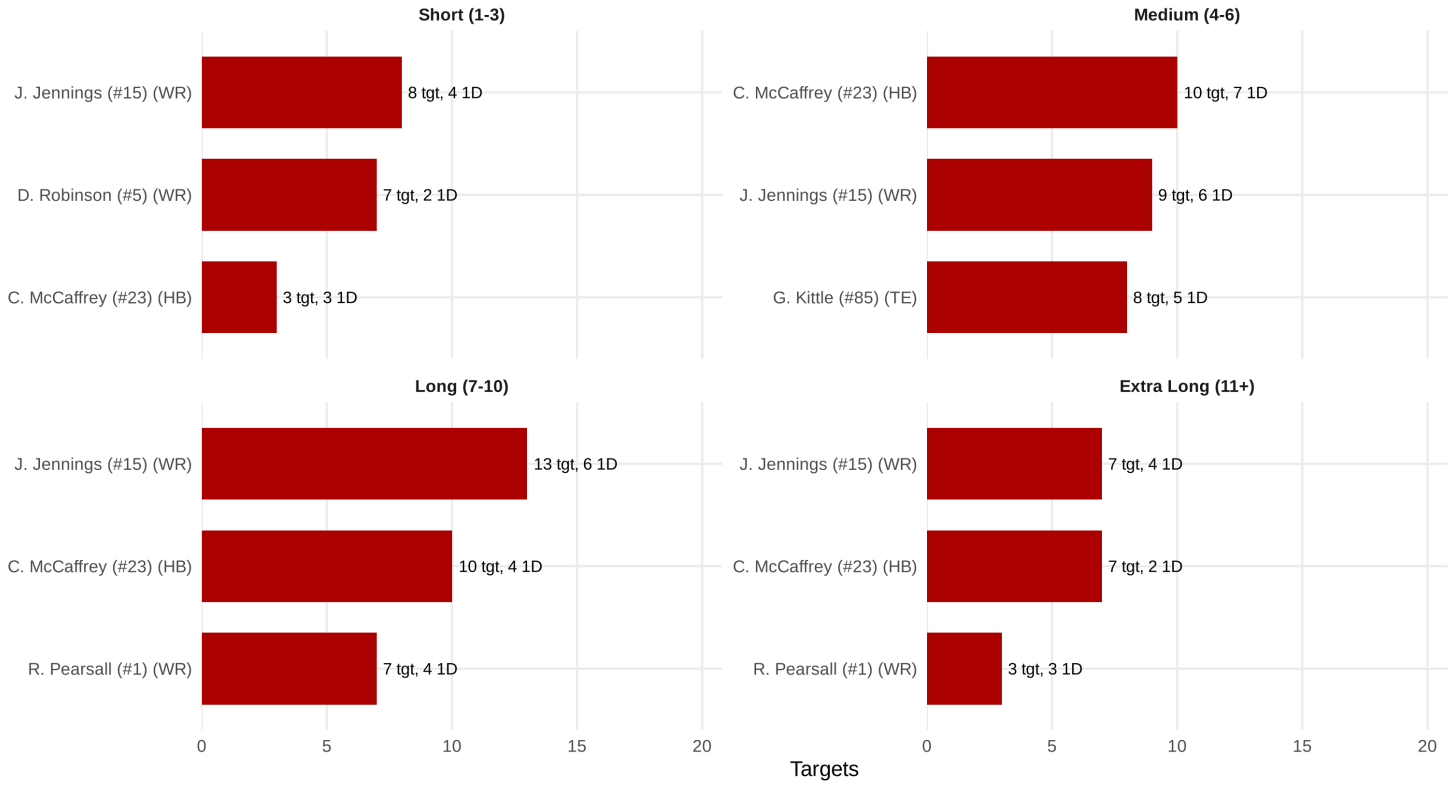
n = number of third-down plays at that distance



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

To go	Plays	Pass %	Conv %	Conv 95% CI	EPA/play	Gun/Pistol %	Avg air yds	Sack %
Short (1-3)	74	43%	64%	52%-74%	+0.16	55%	8.9	3%
Medium (4-6)	63	92%	51%	39%-63%	+0.16	98%	9.4	5%
Long (7-10)	56	95%	45%	32%-58%	+0.39	98%	9.8	6%
Extra Long (11+)	33	91%	30%	17%-47%	+0.46	100%	10.0	3%

Third-down target leaders by distance



1D = first downs gained on those targets. Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

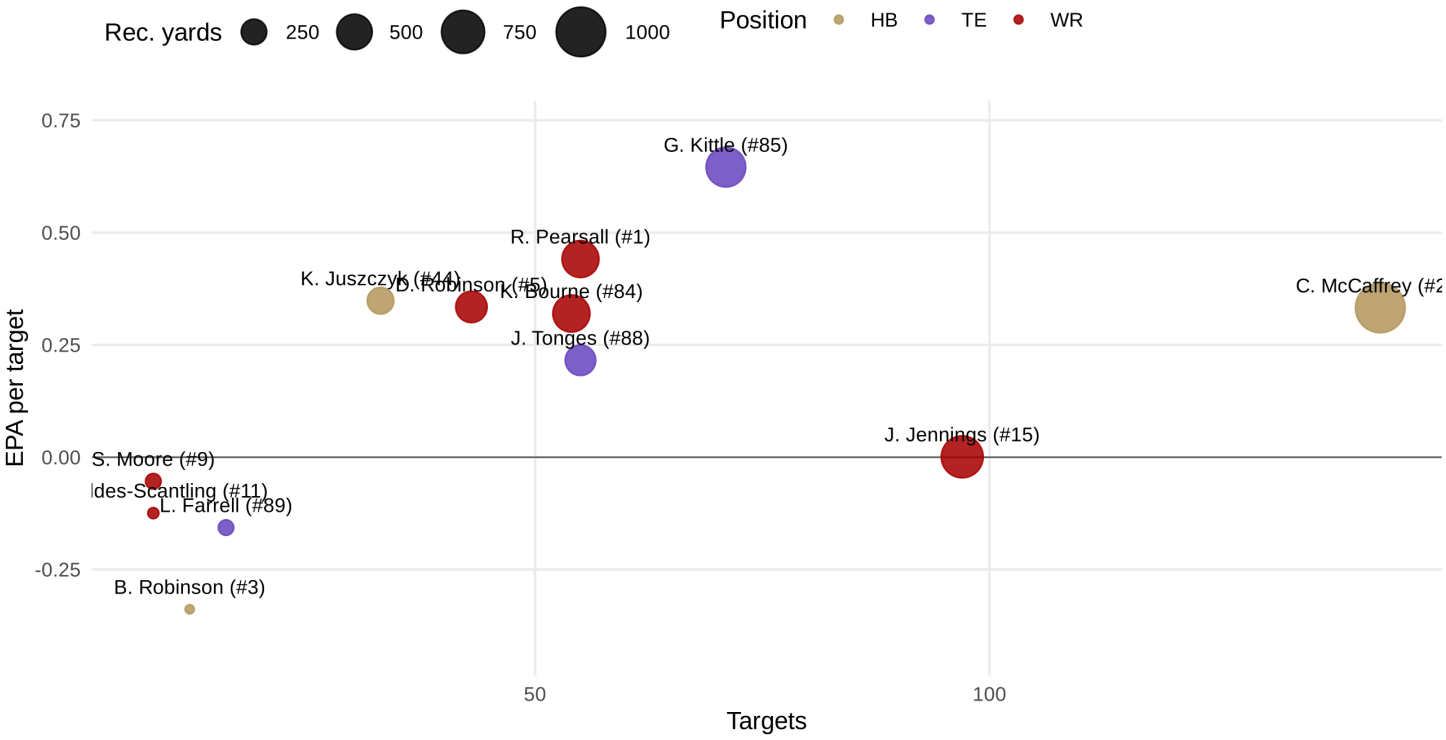
All third downs

Player	Pos	Targets	Share	Rec	Yds	1st downs	EPA/tgt	aDOT
J. Jennings (#15)	WR	37	24%	23	309	20	+0.71	10.6
C. McCaffrey (#23)	HB	30	19%	24	255	16	+0.86	5.2
R. Pearsall (#1)	WR	20	13%	13	184	12	+0.31	15.2
G. Kittle (#85)	TE	18	11%	14	120	10	+0.85	6.4
J. Tonges (#88)	TE	16	10%	11	101	10	+0.48	6.8
D. Robinson (#5)	WR	15	10%	10	118	6	-0.00	15.2
K. Bourne (#84)	WR	13	8%	6	63	4	-0.43	8.6
M. Valdes-Scantling (#11)	WR	3	2%	1	19	1	-0.08	16.3

6. Most effective players and how they produce

Receiving volume vs. efficiency

Players with at least 5 targets

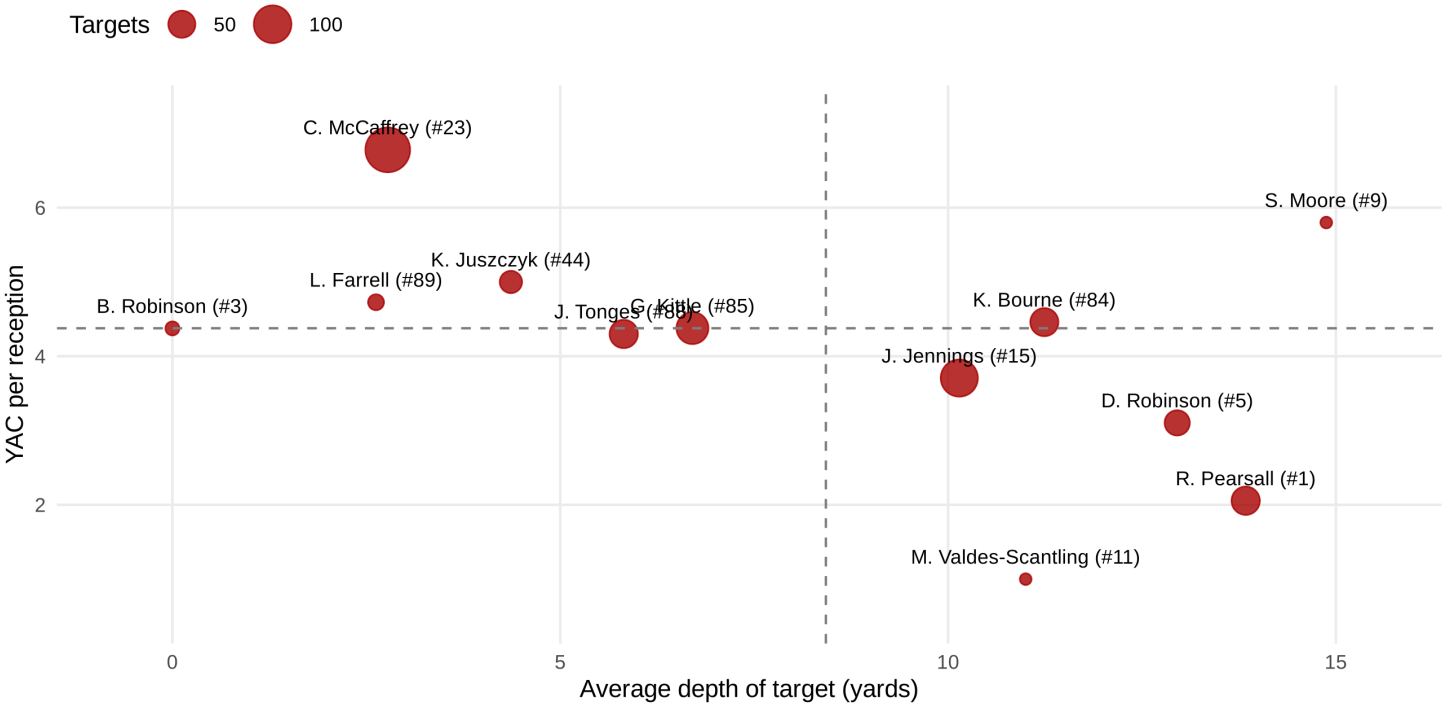


Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Player	Pos	Tgt	Share	Rec	Yds	Catch %	Y/Tgt	aDOT	YAC/Rec	EPA/Tgt	1D	Top route
C. McCaffrey (#23)	HB	143	24%	113	1009	79%	7.1	2.8	6.8	+0.33	55	QUICK OUT
J. Jennings (#15)	WR	97	16%	58	711	60%	7.3	10.1	3.7	+0.00	39	QUICK OUT
G. Kittle (#85)	TE	71	12%	58	634	82%	8.9	6.7	4.4	+0.65	38	QUICK OUT
J. Tonges (#88)	TE	55	9%	40	366	73%	6.7	5.8	4.3	+0.22	27	HITCH/CURL
R. Pearsall (#1)	WR	55	9%	36	548	65%	10.0	13.8	2.1	+0.44	28	IN/DIG
K. Bourne (#84)	WR	54	9%	37	551	69%	10.2	11.2	4.5	+0.32	27	HITCH/CURL
D. Robinson (#5)	WR	43	7%	29	388	67%	9.0	13.0	3.1	+0.33	20	DEEP OUT
K. Juszczyk (#44)	HB	33	5%	29	273	88%	8.3	4.4	5.0	+0.35	13	QUICK OUT
L. Farrell (#89)	TE	16	3%	11	85	69%	5.3	2.6	4.7	-0.16	7	SHALLOW CROSS/DRAG
B. Robinson (#3)	HB	12	2%	8	25	67%	2.1	0.0	4.4	-0.34	1	SWING
M. Valdes-Scantling (#11)	WR	8	1%	4	40	50%	5.0	11.0	1.0	-0.12	1	GO
S. Moore (#9)	WR	8	1%	5	87	62%	10.9	14.9	5.8	-0.05	5	POST

How receivers produce: depth of target vs. yards after catch

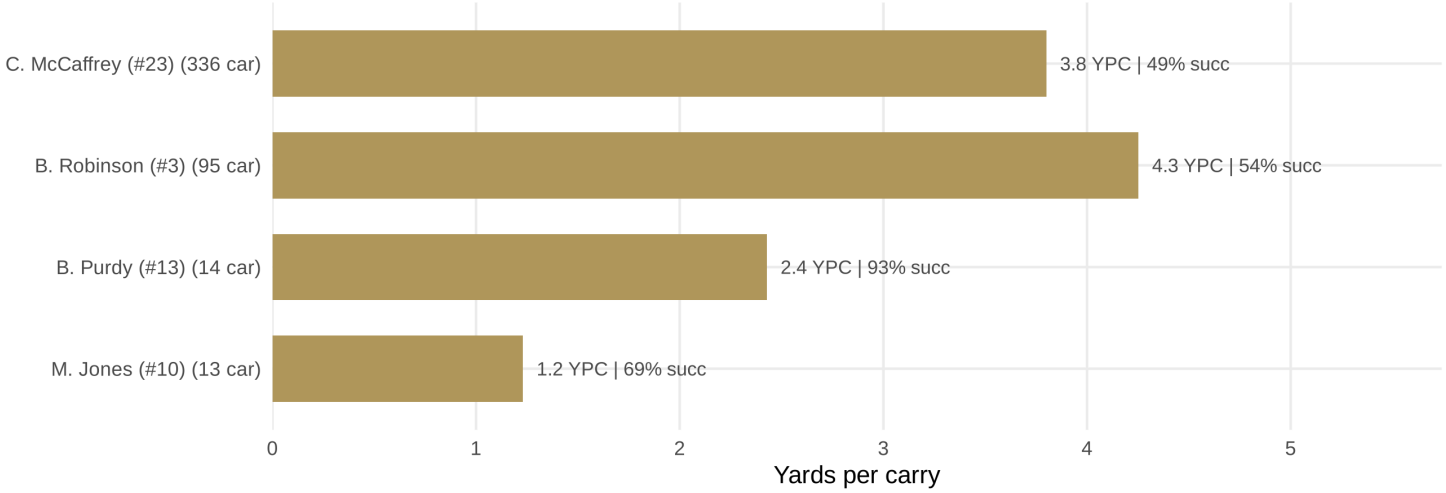
Right = targeted deeper downfield; up = more yards after the catch. Dashed = team medians



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Rushers: yards per carry

Yards after contact is not in this data source, so carries are not split



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Player	Carries	Share	Yds	YPC	Success	EPA/car	YBC	YACo	Explosive %	Stuff %	Top concept
C. McCaffrey (#23)	336	72%	1277	3.8	49%	-0.08	n/a	n/a	8%	19%	Middle
B. Robinson (#3)	95	20%	404	4.3	54%	-0.03	n/a	n/a	11%	14%	Middle
B. Purdy (#13)	14	3%	34	2.4	93%	+0.90	n/a	n/a	0%	7%	Middle
M. Jones (#10)	13	3%	16	1.2	69%	+0.40	n/a	n/a	0%	23%	Middle
K. Juszczyk (#44)	3	1%	-6	-2.0	0%	-2.43	n/a	n/a	0%	100%	Left Guard
S. Moore (#9)	3	1%	14	4.7	67%	+0.18	n/a	n/a	33%	0%	Right End
R. Pearsall (#1)	2	0%	-2	-1.0	0%	-0.84	n/a	n/a	0%	50%	Left End
D. Robinson (#5)	1	0%	6	6.0	100%	+0.32	n/a	n/a	0%	0%	Left End

Total offensive value (rushing + receiving EPA; QB passing EPA is credited to targets):

Player	Carries	Targets	Scrimmage yds	Rush EPA	Rec EPA	Total EPA	EPA/opp
G. Kittle (#85)	1	71	631	-1.0	45.8	44.9	+0.62
R. Pearsall (#1)	2	55	546	-1.7	24.3	22.6	+0.40
C. McCaffrey (#23)	336	143	2286	-26.6	47.5	20.9	+0.04
K. Bourne (#84)	0	54	551	0.0	17.3	17.3	+0.32
D. Robinson (#5)	1	43	394	0.3	14.4	14.7	+0.33
B. Purdy (#13)	14	0	34	12.6	0.0	12.6	+0.90
J. Tonges (#88)	0	55	366	0.0	11.9	11.9	+0.22
M. Jones (#10)	13	0	16	5.2	0.0	5.2	+0.40

7. Passing-game context

Split	Value	Dropbacks	Share	EPA/play	Success	Y/P
Play action	No	536	78%	+0.16	53%	6.9
Play action	Yes	153	22%	+0.09	58%	7.3
Screen	No	647	94%	+0.14	54%	6.9
Screen	Yes	42	6%	+0.27	55%	7.0
Pressure	Clean	516	75%	+0.35	61%	8.3
Pressure	Pressured	173	25%	-0.48	32%	2.9

Targets by depth

Depth	Targets	Share	Comp %	Y/Tgt	EPA/Tgt	INT %
Behind LOS	90	15%	76%	4.7	-0.08	0%
Short (1-9)	315	52%	78%	6.4	+0.22	2%
Intermediate (10-19)	138	23%	66%	11.3	+0.58	6%
Deep (20+)	59	10%	44%	12.4	+0.35	8%

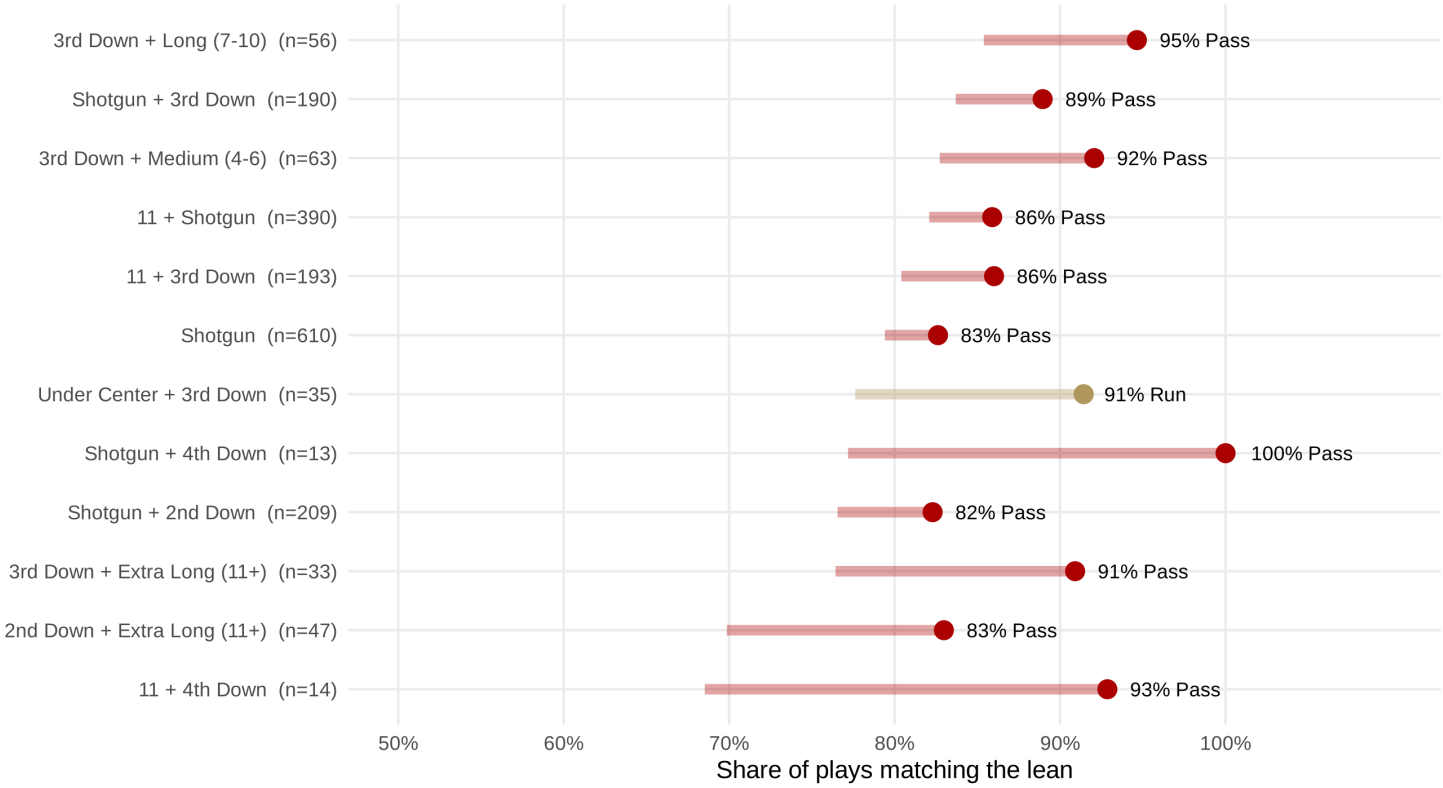
Results by defensive coverage faced (5+ dropbacks)

Coverage	Dropbacks	Share	EPA/play	Success
COVER 3	173	25%	+0.25	62%
COVER 2	166	24%	+0.09	49%
COVER 1	151	22%	+0.06	46%
COVER 4	75	11%	+0.32	56%
COVER 6	48	7%	+0.07	60%
COVER 2 MAN	36	5%	+0.02	61%
COVER 0	20	3%	+0.33	55%
COVER 9	18	3%	-0.18	44%

8. Pre-snap tells and predictability

Strongest pre-snap tells

Dot = observed rate; faded bar runs down to the 95% lower bound; n = plays with that cue



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Always guessing the more common play type is right **59%** of the time. Using personnel, alignment, down and distance bucket (leave-one-out cell rates) is right **74%** of the time, a lift of **15%**.

Methodology and data notes

How the sample was built

Step	Plays
Offensive snaps in file	1259
Run or pass plays (drops penalties / no-plays)	1200
Downs 1-4 (drops PATs / 2-pt tries)	1200
Excluding garbage time	1179
Excluding QB kneels and spikes	1158

WHAT NFLVERSE CANNOT COVER (OR ONLY APPROXIMATES)

- Run concepts: nflverse has no blocking-scheme charting (inside zone, duo, power ...). Section 4 groups runs by gap and direction instead (e.g. 'Left Tackle', 'Middle').
- PFF success codes (pff_OFFSUCCESS): not in nflverse. Success uses the standard 40/60/100 yardage rule only.
- EPA / EP: nflfastR's public model, not PFF's. Values are close in spirit but not identical.
- Yards after contact: not in nflverse, so the before/after-contact split for rushers is unavailable (yards per carry is shown instead).
- Receiver alignment (slot / wide / backfield): not in nflverse. 'Main align' is blank.
- Personnel is counted from the roster positions of the 11 players on the field (RB+FB, TE). A WR listed as TE on the roster counts as a TE. '(*)' = 6+ offensive linemen.
- Garbage time is approximated with win probability (second half, pre-snap WP below 0.01% or above 99.99%). Calibrated against a licensed PFF sample.
- 2-minute flag: inside the final 2:00 of either half (PFF's 4-minute flag has no equivalent).
- Pressure is NGS 'was_pressure' (yes/no); time to pressure is not available.

Column names below refer to the PFF GamePBP layout the nflverse data was mapped onto.

- **Pass play / dropback:** `pff_RUNPASS = "P"`, which includes sacks and QB scrambles. **Run:** `"R"`. Penalties and no-plays (`"X"`) are removed.
- **QB alignment:** Pistol if `pff_PISTOL` is set, otherwise Shotgun if `pff_SHOTGUN` is set, otherwise Under Center.
- **Success (standard):** gaining at least 40% of the distance on 1st down, 60% on 2nd, and 100% on 3rd/4th, or earning a first down. PFF success codes are not in nflverse, so only the standard rule is used.
- **First down:** `pff_FIRSTDOWNGAINED = 1` or yards gained ≥ distance.
- **EPA:** `pff_EXPECTEDPOINTSADDED` as provided. Passing EPA is attributed to the targeted receiver.
- **Target:** a pass play with a recorded receiver, excluding sacks and scrambles. **Receiving yards:** yards gained on completions.
- **Explosive:** runs of 10+ yards, passes of 20+ yards.
- **Field zone:** `ABSOLUTEYARDLINE` is yards to the opponent's end zone (so 1–20 is the red zone).
- **Tell:** a pre-snap cue (personnel, alignment, down, distance, or combinations) with at least 10 plays where one play type reaches 80%. The 95% lower bound uses a Wilson interval.
- **Intervals:** rates show 95% Wilson intervals; run-concept EPA shows ±1 standard error. With samples this small, most differences between groups are not statistically distinguishable; the report shows n everywhere so readers can judge.
- **Garbage time** excluded: TRUE. **2-minute drills** excluded: FALSE. **Personnel "(*)"** marks a jumbo package (an extra offensive lineman replaces a skill player); merged into the base grouping: FALSE.

Data checks:

- Player names from SF_2025_roster.csv (nflverse weekly rosters 2025)

Blank numeric cells (for example yards after catch on runs) are expected and listed for completeness.

Reproducibility: source file MD5 dffa638bb2fd36f893356c7dd959630e. Every table in this report is also saved as CSV and in one Excel workbook next to this file.